

- Description

This module verifies whether a given withdrawal amount is valid based on two conditions:

1. The withdrawal amount **must be greater than zero**.
2. The withdrawal amount **must not exceed the account balance (1000)**.

If both conditions are met, the withdrawal is considered valid and the original balance will be reduced; otherwise, it is rejected.

- Specification

module ATMSystem

var amount: real

 accountBalance = 1000 //initial balance

process isValidWithdrawal(amount: real) flag: boolean

functional scenario form:

 amount <= accountBalance && flag = true and accountBalance =
accountBalance – amount, Valid withdrawal, balance updated, amount meets all
conditions

||

amount > accountBalance && flag = false, Invalid withdrawal, insufficient account
balance

||

amount <= 0 && flag = false, Invalid withdrawal, amount must be greater than zero

end process

end module

- Java Code

```
package code;
public class c19_ATMSystem {
    private double accountBalance = 1000;
    public boolean withdraw(double amount) {
        if (amount > 0 && amount <= accountBalance) {
            accountBalance -= amount;
            System.out.printf("Withdrawal successful! New balance: %.2f\n", accountBalance);
            return true;
        } else {
            System.out.println("Withdrawal failed! Invalid amount or in
sufficient funds.");
            return false;
        }
    }
    public double getBalance() {
        return accountBalance;
    }
    public static void main(String[] args) {
```

```

c19_ATMSystem myATM = new c19_ATMSystem();
    // testcase-VT:
    myATM.withdraw(500);
    myATM.withdraw(600);
    myATM.withdraw(0);
    myATM.withdraw(-200);

    // testcase-FT:
//    myATM.withdraw(-1.7976931348623157E308);
//    myATM.withdraw(1.1476830219940713E296);
//    myATM.withdraw(0);
//    myATM.withdraw(4.169240302727157E298);
}
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c19_ATMSystem.java	100.0 %	4	0	4
✓ c19_ATMSystem	100.0 %	4	0	4
main(String[])	100.0 %	0	0	0
withdraw(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 95.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.4 %	61	16,566	16,627
✓ src	0.4 %	61	16,566	16,627
✓ code	0.4 %	61	16,566	16,627
✓ c19_ATMSystem.java	95.3 %	61	3	64
✓ c19_ATMSystem	95.3 %	61	3	64
main(String[])	100.0 %	21	0	21
withdraw(double)	100.0 %	34	0	34

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.4 %	3	839	842
✓ src	0.4 %	3	839	842
✓ code	0.4 %	3	839	842
✓ c19_ATMSystem.java	75.0 %	3	1	4
✓ c19_ATMSystem	75.0 %	3	1	4
withdraw(double)	75.0 %	3	1	4
main(String[])	100.0 %	0	0	0

5. Fuzz testing instruction coverage: 64.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.2 %	41	16,586	16,627
✓ src	0.2 %	41	16,586	16,627
✓ code	0.2 %	41	16,586	16,627
✓ c19_ATMSystem.java	64.1 %	41	23	64
✓ c19_ATMSystem	64.1 %	41	23	64
withdraw(double)	41.2 %	14	20	34
main(String[])	100.0 %	21	0	21

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c19_ATMSys tem .java	100.0 %	4	0	4
▼ c19_ATMSys tem	100.0 %	4	0	4
● withdraw(double)	100.0 %	4	0	4
● main(String[])	100.0 %	0	0	0

7. Z3 instruction coverage: 95.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	61	22,034	22,095
▼ src	0.3 %	61	22,034	22,095
▼ code	0.3 %	61	22,034	22,095
▼ c19_ATMSys tem .java	95.3 %	61	3	64
▼ c19_ATMSys tem	95.3 %	61	3	64
● main(String[])	100.0 %	21	0	21
● withdraw(double)	100.0 %	34	0	34

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c19_ATMSys tem .java	100.0 %	4	0	4
▼ c19_ATMSys tem	100.0 %	4	0	4
● withdraw(double)	100.0 %	4	0	4
● main(String[])	100.0 %	0	0	0

9. Unimproved V-method instruction coverage: 95.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	61	22,034	22,095
▼ src	0.3 %	61	22,034	22,095
▼ code	0.3 %	61	22,034	22,095
▼ c19_ATMSys tem .java	95.3 %	61	3	64
▼ c19_ATMSys tem	95.3 %	61	3	64
● main(String[])	100.0 %	21	0	21
● withdraw(double)	100.0 %	34	0	34